



# Risk of failure for the salso river railroad steel bridge

Giuseppe Campione<sup>a,\*</sup>, Elio Lo Giudice<sup>b</sup>, Francesco Cannella<sup>a</sup>

<sup>a</sup> Dept. Of Engineering, University of Palermo (Italy), Viale Delle Scienze, 90128, Italy

<sup>b</sup> Dean of DISMART s.r.l. Contrada Andolina, 92024 Canicattì, Agrigento, Italy

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## ABSTRACT

The degradation state of railroad steel bridges is one of the most important and difficult aspects to be interpreted. The use of steel bridges started from 1900, and they are still in service today as railroad bridges. The case study examined was that of a bridge railroad steel bridge having a total length of 112 m and constituted by three isostatic, reticular beams having a Pratt type composition. The substructures (two shoulders and two intermediate piles) are in masonry with a reinforced concrete beam in the upper portion.

The structure (steel bridge and piers) is in a degraded state. To determine the degradation level, non-destructive and partially destructive tests were carried out: - a geometric survey with careful detection of the crack patterns; - a georadar survey on a flat and inclined surface of the pier; - pacometric tests on the upper concrete beams to determine the position and the actual diameter of the reinforcement bars; - endoscopic tests to examine the morphology, typology, and state of conservation of the materials of the piers; - mechanical characterization of masonry constituting the piers through extraction of long core specimens to be tested for compression, chemical and mechanical characterization of the mortar; - microhardness tests for estimating the mechanical tensile strength of the steel constituting the bridge deck. The results obtained were utilized to characterize the strength of the constituent materials for static verification of the steel bridge under permanent and accidental loads, according to the Italian Building Code. The results obtained showed that for this bridge there are no risks of failure, but it was not checked against the Eurocodes, and a retrofitting design is still required.

## 1. Introduction

The territory of Licata (Sicily, Italy) is geographically divided into two by the Salso river, which originates in the Madonie Mountains and flows into the Licata coast. About 500 m from the mouth, the river is crossed by a railroad bridge with an inclination of 19° with respect to the normal direction of the river. Here there is a steel bridge that serves the Licata-Falconara railroad section of the Canicattì-Gela-Comiso line at km 200 + 296 and is located near the Licata railroad station, alongside the Federico II of Svevia road bridge, built on a Morandi project. Fig. 1 shows a plan of the Licata area and the railroad bridge.

This bridge is a steel one erected on the Salso River in Licata (Sicily, Italy) in the 19th century (see Fig. 2) and still present and a living witness of fascinating historical periods and significant events from the past of the territory of Licata. It was built in the last decade of the nineteenth century and solemnly inaugurated on January 1, 1900.

According to the historian Charity, the steel bridge over the Salso began to be considered a reality in 1891 when the government

\* Corresponding author.

E-mail addresses: [giuseppe.campione@unipa.it](mailto:giuseppe.campione@unipa.it) (G. Campione), [elio.logiudice@libero.it](mailto:elio.logiudice@libero.it) (E. Lo Giudice), [francesco.cannella01@unipa.it](mailto:francesco.cannella01@unipa.it) (F. Cannella).

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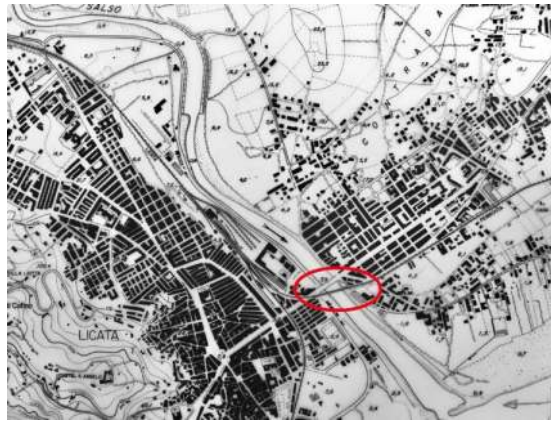


Fig. 1. Plan of Licata territory and the railroad bridge.



Fig. 2. Front view of the Salso railroad steel bridge.

decided to build it, for the rail links between Licata, Terranova (today Gela) and Siracusa. The bridge initially consisted of a multiple lattice metal structure, resting on large masonry piles. The bridge favoured commercial relations with the neighbouring provinces of Siracusa more intimate. Caltanissetta and indirectly facilitated tourist trips to the pretty castle of Falconara.

On November 22, 1915, the bridge was the spectator of one of the greatest catastrophes in the history of Licata. The event, a flood of the Salso river, caused the breakage of the banks and the consequent destruction of a wooden bridge that allowed the pedestrian crossing of the Salso river, located near the railroad bridge which was unharmed thanks to its size and the higher level of the deck compared to the wooden bridge. Fig. 3a) shows the two bridges on the eve of the tragedy, while Fig. 3b) the collapse of the wooden bridge after the disaster. This event claimed 115 victims, who had gathered to browse the wooden structure and who were swallowed up by the impetuous energy of the current. Seven years after the collapse of the wooden bridge, a new one was built on robust piles and according to new techniques that involved extensive use of reinforced concrete. The photo in Fig. 4, taken in 1922, shows the construction phase of the new bridge. In all the photos above, it is possible to note the presence of the steel bridge with the original multiple trellis structure.

The bridge is also found in photos taken in the 1930 s and just before the Second World War. From the beginning of the Second World War to the mid-1950 s, there is no information or photographs referring to the evolution of the two structures. It is only known that in this period of almost ten years, the multiple lattice steel deck was completely replaced with the current isostatic reticular



Fig. 3. a) Front view of the wooden and steel bridges, b) Front view of the steel bridge after the.



Fig. 4. New reinforced concrete bridge.

metal structure; the masonry substructures were still maintained, with the construction of a concrete distribution pier on top of the piles and shoulders. In 1974 the Federico II of Svevia bridge was designed by Prof. Morandi. During its construction, the railroad bridge became an integral part of the construction site, as can be seen from the image in Fig. 5. The steel structure, still in operation today, is at the service of the Italian State Railroads (RFI). This paper is on the field of safety assessment and non destructive testing on steel bridge for railway; which, is a very popular theme in the literature [10,3,8,7,1,6,9].

## 2. Characteristics of the steel bridge

The Salso bridge, as shown in Fig. 6, is a steel trussed bridge divided into three independent spans: the two on the shore have an equal length of 36 m and the central one is 40 m long, giving a total length of 112 m. The constant width of the deck all along the bridge is 4.80 m. The three spans are characterized by steel, isostatic, reticular beams having a Pratt type composition. The substructures (two shoulders and two intermediate piles) are in masonry. Fig. 7 shows the geometry of the truss beam with the cross-sections of the rods making up the truss beams. The upper bridles and the access diagonals of the truss are open sections made up of U-shaped steel elements connected by upper plates and suitable diaphragms (see section 2 of Fig. 7). The lower bridles are formed by steel sections made up of two inverted “T” profiles whose dimensions vary in the longitudinal direction: in the centre of the span (see section 9 of Fig. 7) there is a lower flange having a width of 250 mm and a thickness of 30 mm, section height 380 mm and core thickness 10 mm; the gap between the two coupled beams is 45 mm. At the supports (see section 10 in which the two profiles form an L) having the following dimensions: wing width 100 mm and thickness 8 mm, core width 350 mm and thickness 10 mm and distance between the two profiles 285 mm.

One-third of the way along the span (see section 11), the width of the lower wing is 250 mm and the thickness 20 mm, section height 370 mm, and core thickness 10 mm, while the gap between the two coupled beams is 45 mm.

The lower bridles are connected by cross-beams (see section 12 of Fig. 7) with a double T compound section with the lower and upper bases 310 mm wide, total section height 645 mm, core thickness 10 mm and wing thickness 12 mm. Horizontal braces (section 13), which intersect at the centreline of the crosspiece and have a compound butterfly profile, consisting of 2 coupled  $80 \times 8$  profiles,



Fig. 5. Federico II of Svevia bridge and the Salso railroad steel bridge.



Fig. 6. Lateral view and front view of the Salso railroad steel bridge.

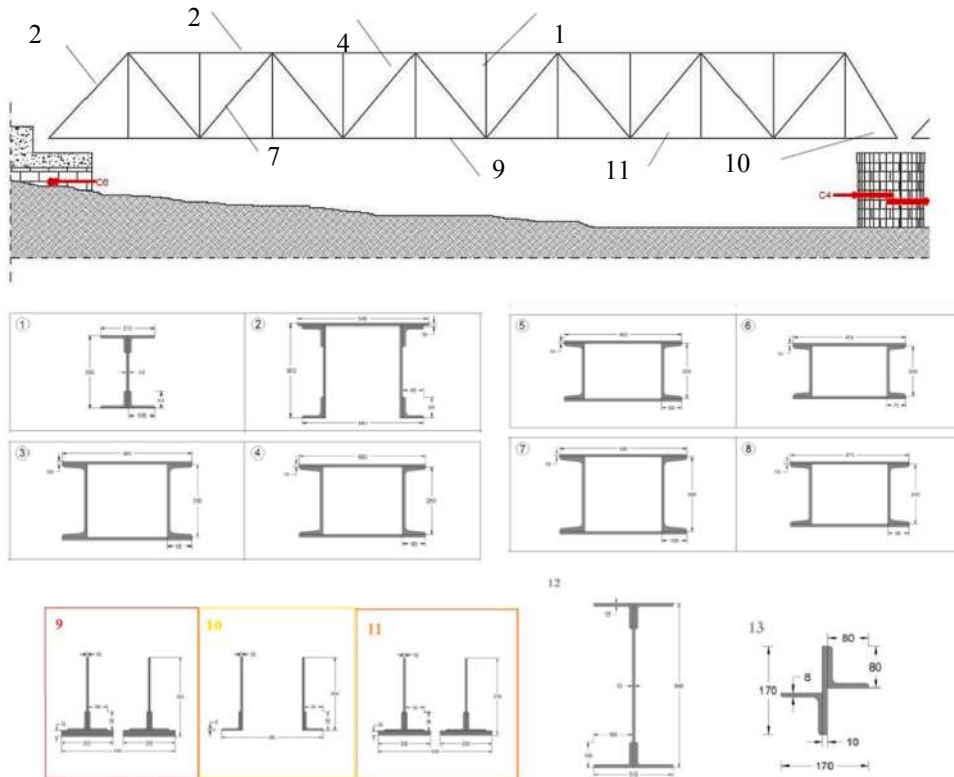


Fig. 7. Half lateral truss with members cross-sections shape.

with a coupling distance of 50 mm. The deck is completed by double “T” profile longitudinal beams (see section 14 of Fig. 7), located 758 mm from the central axis of the bridge, on which the rail track rests. The profile dimensions are the following: the width of the lower and upper bases is 170 mm, the total height is 450 mm, the core thickness 10 mm, the thickness of the lower wing 8 mm, and the thickness of the upper wing 18 mm. The piers (section 1) are all in double T type laminate profile with the following dimensions: lower and upper width 210 mm, total height 280 mm, the thickness of the core and wing 10 mm. The diagonal wall rods have composite laminated profiles, attributable to two coupled UPNs, with dimensions ranging from 200 to 300 mm, respectively, from the centreline to the supports. The elements are tiled. The two spans of the bridge have the same geometry. All the joints between the elements and the different profiles are made by nailing. This connection is made by making the housing holes and then inserting the rivets into the housings and re-riveting with heat, creating a double head, using a particular nailer or riveting machine. From a static point of view, the truss beams are compressed and the diagonal struts stretched. This solution avoids warping by compression of the uprights, which are shorter than the diagonals. Careful visual examination revealed no significant signs of instability on the shoulders, piles or deck. It was observed that almost all metal profiles were affected by the phenomenon of corrosion (see Fig. 8).

The deck is tied to the substructures (piles and shoulders) through steel support devices of fixed type with rocker arm and sliding type with sector rollers which represent, respectively, the “hinge” and “trolley” constraints (See Fig. 9 a). Fixed rocker type appliances (see Fig. 9 b) consist of an upper plate and a lower stand, which touch each other along a generator of two cylindrical surfaces, one concave and the other convex. The stand is made up of a base plate, a central diaphragm, and several symmetrical ribs. Two-sector rollers are inserted under the bottom plate of the rocker arm (see Fig. 9 b). To ensure stability of the device against the horizontal forces normally flowing through the beam, each roller is equipped with a central groove within which the two rolling surfaces are engaged, with limited clearance. The movement of the rollers is guided through vertical strips applied to the heads,



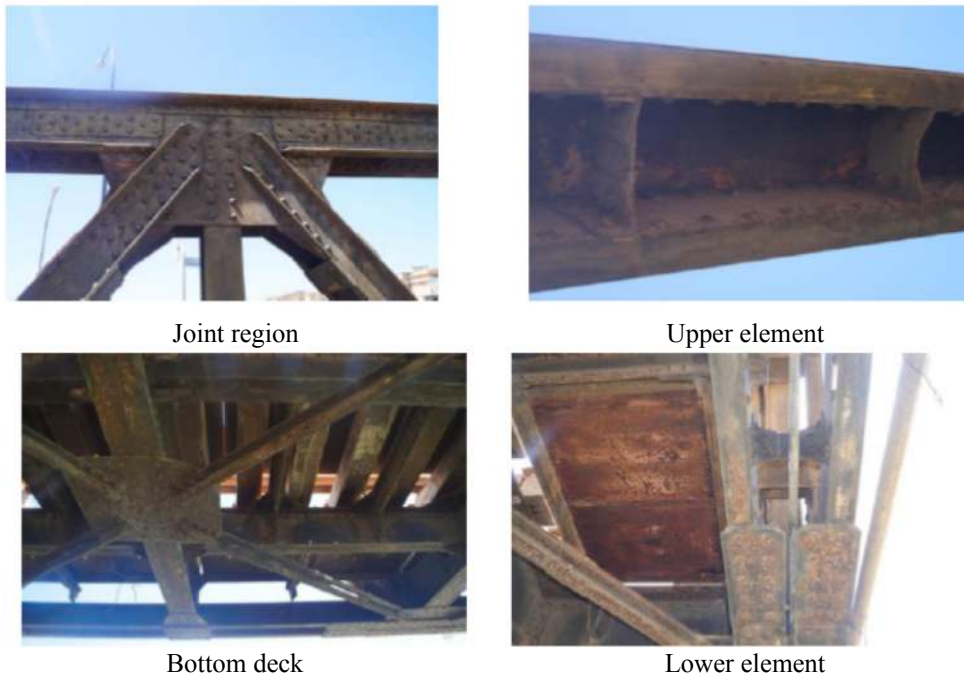


Fig. 8. Typical damages induced by corrosion on steel members of the bridge.



Fig. 9. Damages induced by corrosion on steel devices of the bridge.

which end on both sides with teeth that engage in containment seats.

The substructures constituted by two shoulders and two piles are realized in masonry with squared stone blocks having at the top a partition in reinforced concrete on which the support devices rest (see Fig. 10). The shoulders have a height of 340 cm; one of the stacks of 396 cm and the other 530 cm, and their axis is inclined by  $19^\circ$  with respect to the direction normal to the bridge axis (they follow the direction of the river's flow). It has to be stressed that the substructures are those of the original bridge (with multiple trellises); they were maintained for installation of the new deck. About the reinforced concrete curb, it is therefore assumed that it was made when the steel bridge was replaced. Fig. 11.

Signs of deterioration are also visible on the piles, where there are lesions, loss of concrete cover, and disintegration of the mortar joints.

Steel began to be used for deck construction with lattice or multiple lattice structures from the second half of the 19th century. The substructures, imposing large stacks, were built in square stone masonry for the curtain wall and loose stone masonry for filling.



Fig. 10. Masonry central pier.



Fig. 11. Damages due to: a) concrete cover spalling; b) disintegration of mortar joints.

The foundations were made with beaten wooden poles. The latter were obtained from tree trunks, with a tip diameter of 10 to 20 cm. The most widely used woods were red larch, pine, oak, elm and chestnut. The richer woods are in resin, the more resistant and longer-lasting they are. Therefore nordic timber is the best (Russian, Swedish, and Norwegian pine). The fibers must be straight and the stem, debarked, free of knots and impurities. At the pointed end, a cast iron or steel tip is applied, and a steel ferrule to the head, which during insertion into the ground receives the blows of the mallet. During insertion, they can be damaged. Poles subjected to the alternation of dry and wet (variation of the level of the water table) deteriorate rapidly. By contrast, if always immersed in water, they keep much longer. These construction techniques were widely used for the construction of several railroad bridges throughout Italy, as can be seen from the photographs below.

### 3. Experimental program

The investigation program carried out for the railroad bridge over the Salso River regarded the following: - a complete geometric survey with careful detection of the crack picture; - a georadar survey on a flat and inclined surface for a total of 47 m, making it possible to examine the bridge and determine the presence of any cavities, air inclusions or other defects; 3 pacometric tests extended to 1 m<sup>2</sup> each to determine the position of the reinforcement bars with consequent diameter and thickness of the concrete cover of the shoulder curb; - 6 endoscopies to examine the morphology, typology, and state of conservation of the materials. For characterization of the materials, the following investigations were carried out: - six 150 cm long cores from which 18 cores were derived for crushing tests on concrete and masonry elements; - 18 tests for chemical and mechanical characterization of the mortar; - 54 microhardness tests for estimating the mechanical tensile strength of the steel constituting the beam and components of the bridge deck.

No specific measurement and evaluation of corrosion losses were estimated because it was not possible to extract steel coupons from steel bridge because it was considered as historical monuments. Moreover, it was not possible to perform non-destructive measurements of the real thickness of the cross-sections.

Visual inspections shows that corrosion processes are in act but it was not possible to estimate the loss of mass.

### 4. Experimental investigation on steel beams of the deck

[Autor]

The mechanical characteristics of the steel of the deck carpentry were determined utilizing in-situ microhardness tests. It has to be stressed that hardness testing is an extensively used form of materials test. It is comparatively easy to perform and is typically completely or minimally non-destructive, and also most of the instrumentation is inexpensive when compared to other types of material verification equipment. Furthermore, it can typically be carried out directly on the component without major alteration. The information from a hardness test can be considered complementary and frequently be used in tandem with other material verification techniques such as tensile to steel coupons extract from structural elements resulting the latter the best to determine yielding stress of material but to be considered an invasive testing technique not always to be utilised in historical structures. Because indirect tensile tests based on hardness tests it was a risky to base the static assessment only on hardness measurements and therefore static verification for stress level can be considered qualitative and as a preliminary check to more detailed control.

The microhardness tests with the Leeb method, was based on the dynamic (rebound) method according to the following methodology: a stop body with a metal tip hard is projected by a spring against the surface of the object to be analyzed; the impact causes slight deformation of the surface, which leads to a loss of kinetic energy. This loss is calculated by measuring the speed of impact ( $v_i$ ) and rebound ( $v_r$ ) at a precise distance from the surface. These are measured through a permanent magnet in the impact body, which generates an induced voltage in the firing pin probe. The voltage identified is proportional to the speed of the impact body. Signal processing provides hardness. Fig. 12 showed the execution of tests. Table 1 gives the Leeb Hardness values, which in accordance with ASTM A956 was related to Vickers hardness and yielding stress. Hardness values were related as suggested in ASTM A956 to range of yielding stress in between 350 and 400 MPa.

According to the Italian code for existing structures the design values for steel profiles has to be assumed as



Fig. 12. Hardness test of steel profiles with Leeb method.

**Table 1**  
Leeb test results.

| position          | Leeb Hardness(HLD) | Vickers Hardness (HD) | Yielding stress(MPa) |
|-------------------|--------------------|-----------------------|----------------------|
| Beam - span 1     | 367                | 122                   | 414                  |
| Beam - span 2     | 374                | 128                   | 392                  |
| Beam - span 3     | 378                | 131                   | 440                  |
| Diagonal - span 1 | 378                | 131                   | 440                  |
| Diagonal - span 2 | 372                | 126                   | 426                  |
| Diagonal - span 3 | 369                | 124                   | 420                  |

$$f_{yd} = \frac{f_y}{\gamma_s \cdot \gamma_c} \quad (1)$$

$\gamma_s$  being the safety factor of the material and  $\gamma_c$  the corrective coefficient depending on the confidence level reached with the tests carried out. According to Italian code (2018) depending on the number of tests carried out for the type of structures investigated the  $\gamma_c$  was 1.20, while  $\gamma_s = 1.05$  according to Eurocode 3 [5]. Therefore, from Eq. (1) the allowable stress value was 334 MPa. In addition, pacometric investigations were carried out according to on three different areas of the concrete beams on the pier for an extension of one square meter with a 20 cm mesh lattice as prescribed by the rules on the curb placed at the top of the pile where the constraints of the metal beams are housed. The investigations aimed to make it possible to detect horizontal reinforcements with a diameter of 14 mm, and vertical bars with a diameter of 22 mm. In accordance with the date of construction, the bars are of the smooth type. The concrete cover was 3 cm. No degradation of steel bars with corrosion was observed.

## 5. Compressive strength of masonry abutment

The compressive strength of masonry, as is well known, depends on the resistance of the stone elements, on the mortar, and the wall texture. Table 2 gives the values of weight density and compressive strength for each specimen investigated. In Table 2 the specimens were named with letters in order of depth from the outermost (letter A) to the innermost (letter F). The variance of compressive strength values was large perhaps for the heterogeneity of material adopted to construct masonry piers. Generally, compressive strength can be deduced according to two ways: – inductively, deducing it from the mechanical characteristics of the components, also taking into account the texture; - experimentally through tests with double flat jacks. However, the latter procedure is not recommended in the case in question due to the following circumstances: the stone blocks have such dimensions that the limitations provided for by the ASTM C 1197 standard cannot be respected; in the majority of cases the cutting depth (about 260 mm), and therefore the area of influence of the jack, affect areas of masonry with remarkable differences in behaviour (the external facing, of limited thickness, is generally made with squared stone with high mechanical characteristics); mortar joints are often missing in the masonry because they are strongly degraded by atmospheric agents; the test with a double flat jack is generally aimed at determining the elastic modulus of the ashlar and it is not certain that the breaking strength can be determined with it. For determination of the mechanical strength of the masonry, the first procedure was chosen. For this purpose, the average compressive strength of the stone elements,  $f_{bm}$ , was determined using a compression test on the extracted samples as per UNI EN UNI EN [11], from which the characteristic compressive strength of the  $f_{bk}$  element is subsequently obtained through the relationship  $f_{bk} = 0.80$

**Table 2**  
Compressive strength of extracted core.

| Carrot extracted | Specimen | Compressive strength (MPa) | Weight density(daN/m <sup>3</sup> ) |
|------------------|----------|----------------------------|-------------------------------------|
| C1               | A        | 12.70                      | 2248                                |
| C1               | E        | 2.50                       | 1725                                |
| C1               | F        | 2.90                       | 1894                                |
| C2               | A1       | 28.30                      | 2689                                |
| C2               | A2       | 16.80                      | 2653                                |
| C2               | C        | 10.20                      | 2290                                |
| C3               | A1       | 9.70                       | 2196                                |
| C3               | A2       | 10.60                      | 2240                                |
| C3               | C        | 7.30                       | 1960                                |
| C4               | A1       | 11.30                      | 2198                                |
| C4               | A2       | 7.50                       | 2191                                |
| C4               | D        | 5.10                       | 2097                                |
| C5               | A        | 12.10                      | 2229                                |
| C5               | A1       | 15.60                      | 2266                                |
| C5               | D        | 3.70                       | 1968                                |
| C6               | A        | 2.00                       | 1866                                |
| C6               | A1       | 3.40                       | 1952                                |
| C6               | D        | 10.60                      | 1908                                |



**Table 3**  
Compressive strength of masonry in MPa.

| Elements                  | Stone | Mortar | Masonry |
|---------------------------|-------|--------|---------|
| External masonry pile     | 9.77  | 5.45   | 4.3     |
| Internal masonry pile     | 4.23  | 5.42   | 2.8     |
| External masonry shoulder | 7.00  | 5.40   | 3.60    |
| Internal masonry shoulder | 2.30  | 5.39   | 1.80    |

$f_{bm}$  (chap. 11.10.3.2.1). Subsequently, the resistance of the mortar in-situ is estimated through non-destructive techniques, and the strength of the masonry is indirectly determined as a function of the resistance of the mortar and the stone. [Table 3](#).

## 6. Core drilling, endoscopic tests and compressive strength of core drilled

Coring on the piles was carried out in the positions indicated in [Fig. 13](#) and was aimed at characterizing the morphology of the masonry itself, by investigating the internal structure and the constituent materials (types of mortar, brick, stone; thickness and conformation of the wall facings; other details, etc.) and also at characterizing the mechanical strength of the stone constituting the shoulder.

In particular, the thickness of the outer face (curtain) of about 60 cm was deduced from the extracted carrots. Subsequently, the carrots were sampled to extract portions which, suitably worked (sanding/capping), were subjected to compression tests with the aim of determining the value of the compressive strength of the block.

Carrots 1, 2, 3, 4, 5, shown in [Fig. 14](#) a), b), c), d), are extracted neatly and placed in the appropriate boxes, from which it is possible to identify the succession of the layers crossed and therefore the relative depth.

The endoscopic investigation was performed adopting the equipment shown in [Fig. 15](#), inserting a small endoscopic probe in holes at least 20 mm in diameter. The results of the visual inspection can be recorded through a video recording system, in images or video files. [Fig. 16](#).

[Fig. 15](#) shows the images of the endoscopic investigations performed with an MIGS v03 handheld model endoscope. The endoscopic images were taken inside each coring for a depth of about 160 cm. The entire wall facing is uniform, and there are no obvious discontinuities in the wall type. However, as can be seen in the images, there are slight cavities inside the masonry. Furthermore, the part that constitutes the curtain and that of the filling are clearly distinguished.

Adjusted and reduced to the standard size, the extracted carrots were crushed according to the UNI EN 12,390 standards.

## 7. Georadar investigation

The method commonly called Georadar (Ground Probing Radar - GPR) is an investigation system based on reflection of electromagnetic waves (ASTM Standards [2]. Operationally, it consists in sending high frequency electromagnetic impulses (between 10 and 2000 MHz) and in measuring the time taken by the signal to return to the receiver after being reflected by any discontinuities intercepted during its path; these reflections are generated at the contact surfaces between materials with different dielectric constant or different electrical conductivity, changing the electrical properties of the material. The instrumentation used (RIS MF Hi-Mod) consists of a structure where it is possible to install the antenna body, the control unit, the encoder and finally a laptop or tablet to view the scans carried out in real time. The antenna body is made up of a box containing the 900/2000 MHz configuration. The modular configuration makes it possible to assemble multiple antennas by means of a chain link having the possibility of simultaneously scanning more or less large surfaces. An IDS Multichannel FW control unit allows you to manage all types of antennas. The antenna is slid along one or more investigation lines, according to an established grid, so that the signals can subsequently be associated with the real position from which they were emitted and received. Data processing takes place with specific software in two distinct phases: transformation of raw data through filtering and processing algorithms, so as to produce clearly legible data; interpretation of the data to identify the signals (targets) attributable to cavities, artefacts and structures and to estimate their depth. The depth of the reflection surfaces, identifiable on the radar sections (radargrams), is determined using a mathematical relationship that links the aforementioned depth to the reflection time, directly readable on the radar sections, through the speed of the radar pulses in the investigated material ( $V_m$ ). The average speed values of the radar pulses in the different types of material can be obtained from the technical literature, but it is preferable to use the appropriate speed analysis algorithms with which all GPR data processing software are equipped. For the same material, the speed of the radar signal can in fact also vary significantly according to parameters such as porosity and humidity.

[Fig. 17](#) shows a phase of the test with georadar on the wall facing of the bridge shoulder.



**Fig. 13.** Position of extracted core.



Fig. 14. Core extracted from piers and batimens.



Fig. 15. Endoscopy for investigation on masonry.

From the georadar survey carried out, no anomalies were found inside the masonry. In some radargrams shown in Fig. 18, perturbations were found, probably attributable to large-sized stone material. Below is one of the radargrams of the survey on which the perturbations are highlighted in the red boxes.

For determination of the mortar strength, a penetrometric test was used, which makes it possible to estimate its compressive strength by measuring the energy dissipated by a special drill in making a hole with a standard size. Specifically, this is an indirect technique based on micro-destruction of the material by means of a portable drill equipped with a tip with a tungsten cutting edge and connected to a control unit that records the energy spent to drill a hole in the mortar joint. The equipment used consists of a PNT-D 662 series portable drill with a 4 mm diameter bit.

These tests were performed through laboratory analyses on samples extracted in-situ. Selection in-situ can be done from the joints or with small cracks made in the masonry. It is necessary to carefully evaluate the removal of material from carrots, in the case of coring, paying particular attention to taking samples as undisturbed as possible with respect to the cut made and not showing excessive washout. The size of the samples to be taken can also be very small (in the order of one centimeter). The equipment used is shown in Fig. 19.

The test procedure involves the execution of six holes for a depth of 14 mm; for each of them the energy used is recorded, from which the resistance of the mortar is ascertained by means of the experimental law

$$\sigma = 0.0497 \cdot E^{-2.7619} \quad (\text{MPa}) \quad (2)$$

with  $\sigma$  the measured tension of the mortar, and  $E$  the energy expended by the drill. The tests carried out gave an average stress of 5.45 MPa. Based on these values, the mortar could be classified as M5 according to the Italian Code (2018). The characteristic strength of the blocks for the external part of the piles, the internal part of the piles, the external part of the shoulders and the internal part of the shoulders was 9.77 MPa, 4.23 MPa, 7 MPa and 2.27 MPa. This gave a compressive strength of the masonry of 4.3 MPa, 2.8 MPa, 3.6 MPa and 1.8 MPa for the external part of the piles, the internal part of the piles, the external part of the shoulders and the internal part of the shoulders, respectively.

## a) Pier 1



## a) Pier 2



Fig. 16. Endoscopic tests on hole determined on per by extraction of core 1 and 2.



Fig. 17. Georadar test on pier 1 and 2.

## 8. Static verification

For static verification of the railroad bridge, before 1916 and before unification of the railroad networks in Italy, reference was made to two design trains called category IV and category V trains. After 1916, with the entry into force of the technical standards, public railroads included two categories called A main lines, on which there were heavier locomotives, and category B normal gauge lines, not included in category A, and narrow-gauge lines. For category A bridges, three 180 kN isolated axes 1.5 m apart or 210 kN isolated axis were to be considered. The limit stresses of the material were variable according to the length of the element.

The actions included were according to Eurocode 1 [4]: - own weights; - the permanent loads carried out by the bridge; - the vertical variable actions with reference to the load model LM71, SW / 0 and C3 envisaged by the Italian standards and by the RFI



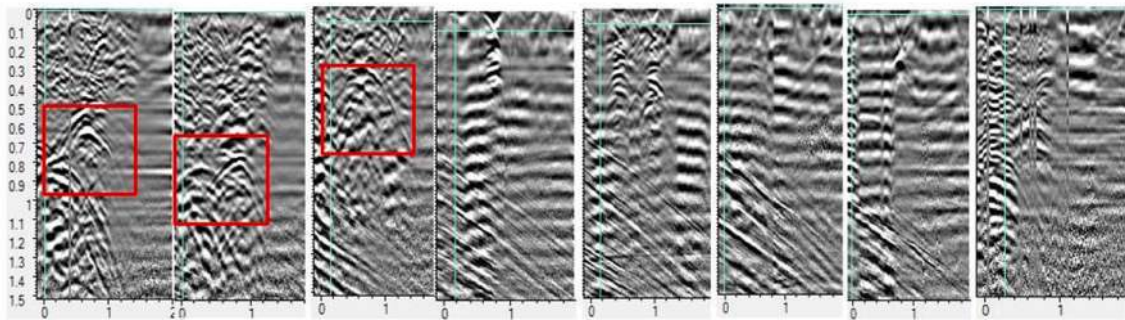


Fig. 18. Recorded radargrams on pier 1 and 2.



Fig. 19. Test set-up for compressive mortar characterization.

Design Manual; - the hydrodynamic actions due to the river. The variable vertical actions simulated the passage of the train considered as a traveling load. The actions generated by the latter were broken down into concentrated and / or distributed loads. The positioning of the load was chosen to maximize the effects in terms of stresses and displacements by providing for 6 different positions for each type of variable load. The calculation model adopted for finite elements was that of Fig. 20, in which the rods of the truss beams have been modelled with beam type elements and the masonry stacks and the concrete curbs with brick type elements.

The static effects produced by normal rail traffic are incorporated in the model; the load diagram for the LM71 and SW / 0 models are summarized in Fig. 21. An adaptation factor of 1.1 was applied to the models. Fig. 22.

Among the horizontal variable loads, lateral action (meandering) and starting and braking actions were considered. Winding was considered as a concentrated force acting horizontally, applied to the top of the highest rail, perpendicular to the axis of the track. The characteristic value of this force is 100 kN. This lateral force must always be combined with the vertical loads, that is with the different load models, and also multiplied by the adaptation coefficient “ $\alpha$ ” referring to the load model considered. The lateral action was studied for the LM71 and SW/0 models in the 6 positions.

The braking and starting forces act on the top of the track. These horizontal forces are considered uniformly distributed over a track length  $L$ , determined to obtain the most severe effect on the structural element considered. Braking and starting actions must also be studied by combining them with vertical actions, and therefore with the load models considered and in the various positions.

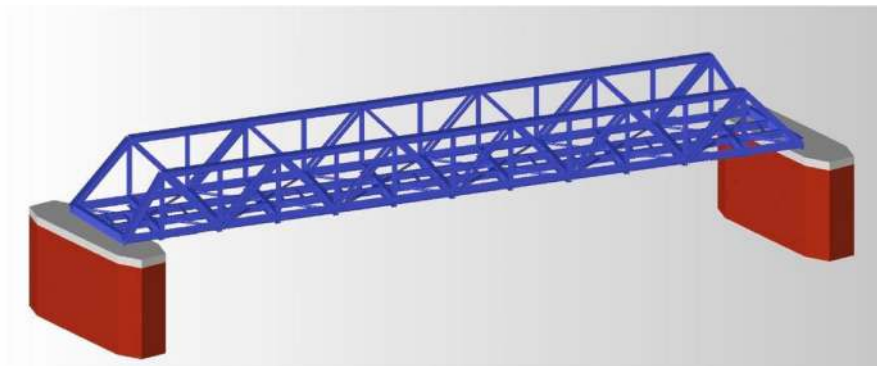


Fig. 20. FEM model examined.



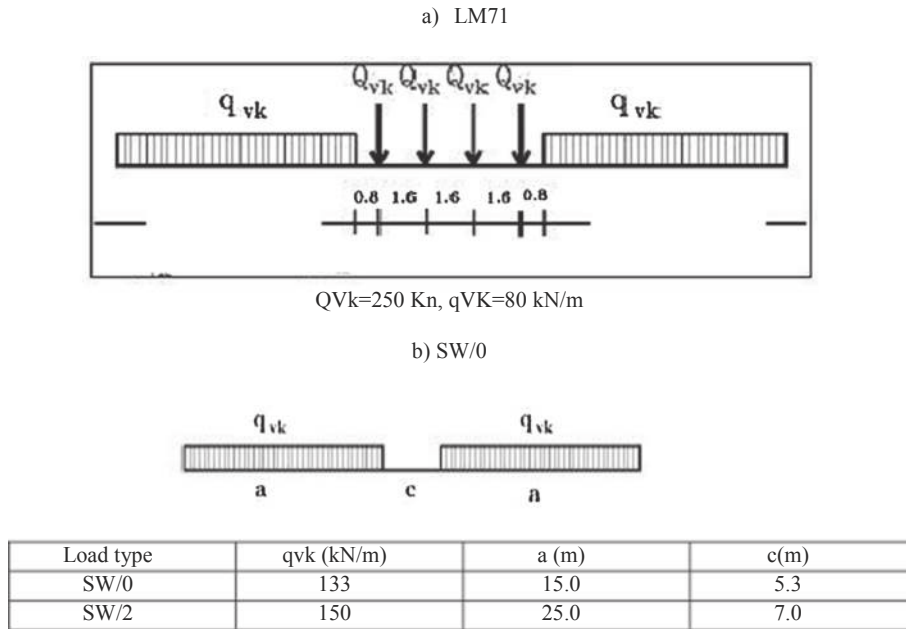


Fig. 21. Loading scheme adopted for train action: a) LM71; b) SW/0.

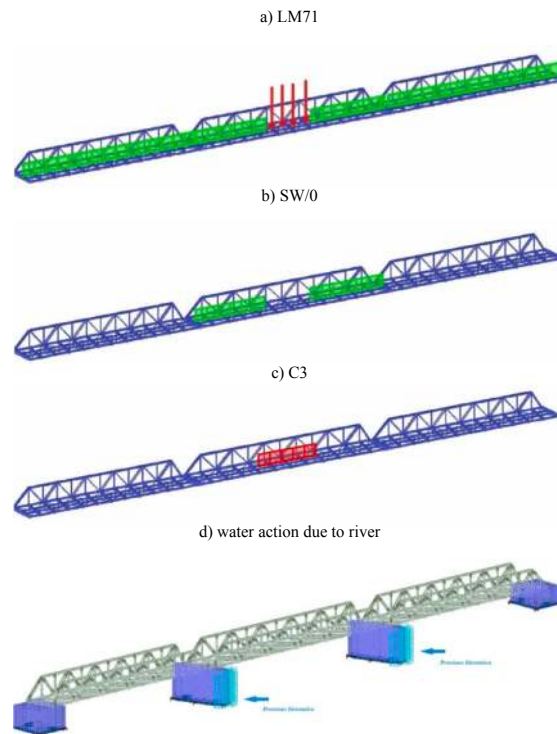
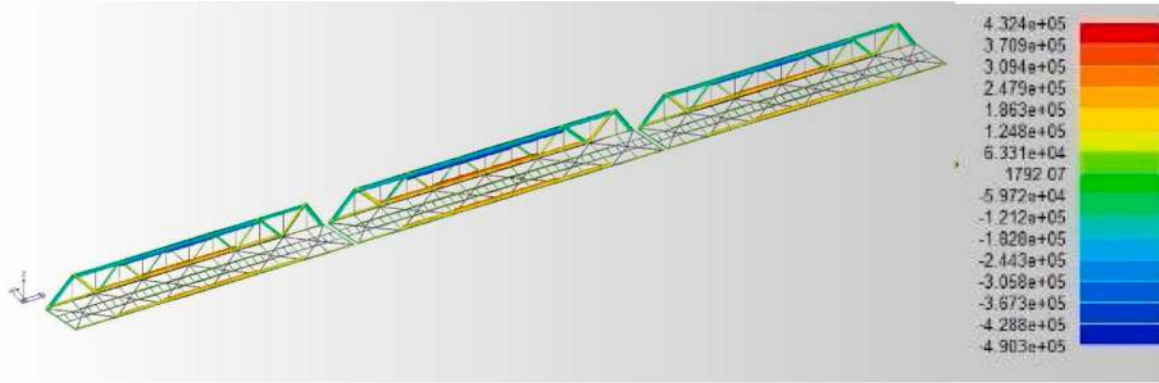


Fig. 22. FEM model with action due to: a) LM71; b) SW/0; c) C3; d) hydraulic action on piers.

The characteristic values of the two actions are starting 33 kN/m and braking 20 kN/m. Fig. 19 shows the finite element numerical model for one of the 6 configurations examined for the different loads.

Adopting a simply supported beam model subjected to a dead load of 40 kN/m and a permanent load of 10 kN/m, simple hand calculation gave the following maximum bending moment and shear force:

a) envelope of axial forces (1000 daN=100 kN)



b) envelope of bending moment (1000 daN cm=1 kNm)

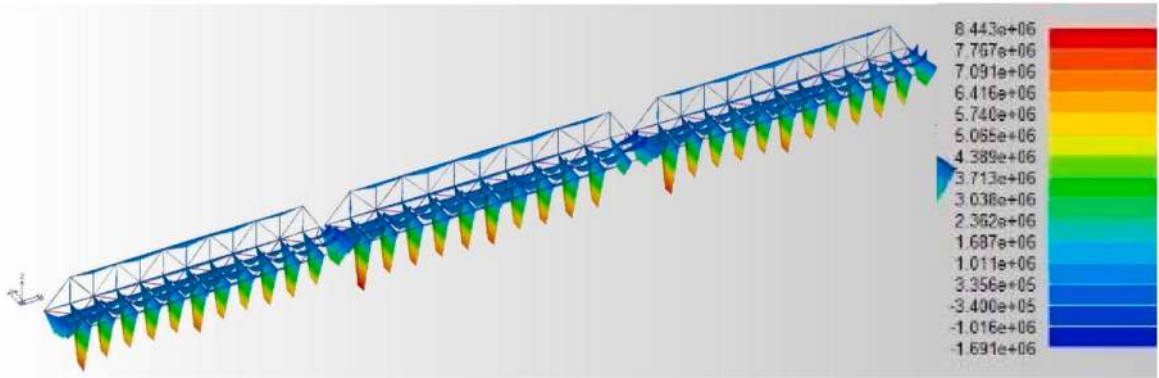


Fig. 23. Envelope of a) axial forces; b) bending moments.

$$M_{sd} = 1.09 \cdot 1.1 \cdot 1.45 \cdot \left( \frac{4 \cdot P_{1k} \cdot L}{2} + \frac{(q_{1k} + q) \cdot L^2}{8} \right) = 57772 \text{ kNm} \quad (3)$$

$$V_{sd} = 1.09 \cdot 1.1 \cdot 1.45 \cdot \left( \frac{4 \cdot P_{1k}}{2} + \frac{q_{1k} \cdot L}{2} \right) = 4908 \text{ kN} \quad (4)$$

The axial force on the upper beam results from dividing the moment  $M_{sd}$  deduced from the beam diagram by the distance between the beams:

$$N_{sd} = \frac{M}{2 \cdot h} = 8665 \text{ kN} \quad (5)$$

The moment on the lower current induced by the distributed load is equal to

$$M_{sd} = 1.09 \cdot 1.1 \cdot 1.45 \cdot \frac{(q_{1k} + q) \cdot L^2}{8} = 126 \text{ kNm} \quad (6)$$

The stresses and displacements, determined on the bridge structures by static application of the load models, were increased to take into account the dynamic nature of the transit of the trains. The dynamic application coefficient of 1.09 was adopted as required by the standards, the first flexural frequency typical of the structure being 5.80 Hz, and the corresponding length of the span  $L_\phi$  40.0 m.

For the piles, in addition to the action transmitted by the deck, hydraulic thrust was also considered. This action was calculated considering the maximum water speed of 7 m/s, which leads to a thrust of 2500 daN/m<sup>2</sup>. Fig. 23 shows the thrust of the water on the two piles in blue. For security verification, reference was made to the fundamental combination in the ULS and SLS, which is rare. The stresses were calculated with linear elastic analysis. Fig. 24 shows the diagrams of normal stresses and bending moments for the envelope of the elementary load conditions combined according to the S.L.U. criteria.

Checks on the S.L.U. for steel structures are divided into resistance checks, local stability checks, and global instability checks.

For verification of the resistance, the interaction M–N domain was calculated according to Eurocode 3, under the hypothesis of

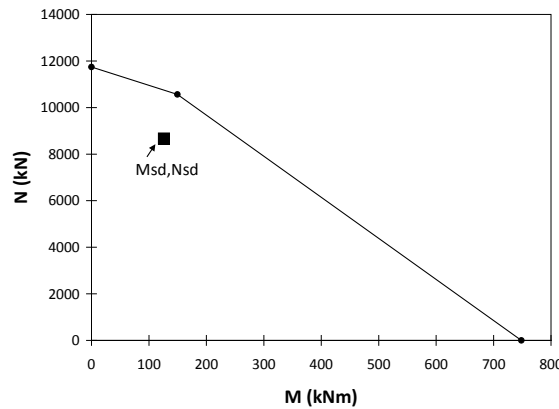


Fig. 24. Bending moment-axial force domain for steel profile type 2.

plastic behaviour of the steel. The calculation resistance was obtained from the yield strength equal to 375 MPa divided by the safety factor 1.05 and the confidence factor that takes into account the type and number equal to 1.35.

The domain, can be approximated with two straight lines: the first passes through the point corresponding to compression only and the other to flexure only. The intermediate point has coordinates 0.2 N, 0.9 M if the double T profile is loaded along the strong axis and 0.5 N, 0.9 M if it loaded along to the weak axis. In this case, the reference diagram is related to the verification of the superior current, which it is the most stressed member.

Fig. 24 shows the domain of interaction with the stress calculation. Since the point with coordinates  $M_{sd}$  and  $N_{sd}$  is internal to the domain, the verification is satisfied. For the determination of the stability of the member, since the slenderness is reduced, there is no reduction in plastic strength.

Concerning global verification of instability through the SAP2000 software, a buckling analysis was carried out, which provided for the heaviest load combination with a collapse multiplier of 9.91, so it is possible to determine that the structure was safe against this phenomenon.

The deflections checks for the serviceability limit states, as suggested by the railroad design manual, consist in calculation of the arrow in the presence of train LM71 or SW/0; according to Italian Design Law, the arrow increased by the dynamic factor of 1.1 has to be less than 1/600 of the length of the span. Furthermore, for the comfort of passengers, the limit is further reduced to 1/1000 of the length of the span. Displacements in the negative (downward) z-direction of all load combinations were examined. Of these, the one with the maximum negative vertical displacement, which in this case corresponds to the minimum, is the combination number 5 that corresponds to the load model LM71, with a maximum displacement of 12.34 cm, which is higher than the limit equal to  $1/600 \times 4000 = 6.66$  cm, and the verification is not satisfied. Static verifications are not satisfied either in terms of strength or in terms of deformability.

For masonry piles, a bending test must be performed. The condition of pressure break in the plane is associated with crushing of the masonry at the compressed edge of the extreme sections. For low values of axial action N, the extension of the compressed area is modest, and a large flexure crack is raised. The wall does not develop a kinematic tipping mechanism similar to that of a rigid block. The analysis of masonry in flexure and axial force is simplified in a stress block, shown in Fig. 25.

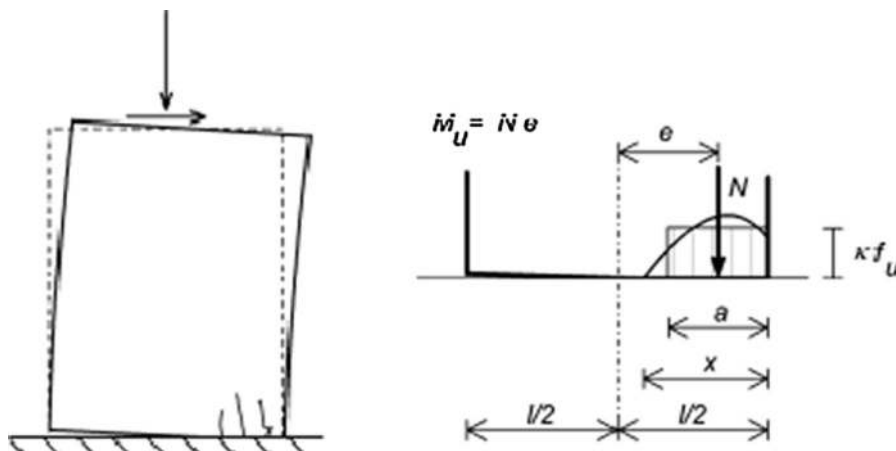


Fig. 25. Flexure-axial force failure mechanism of masonry.

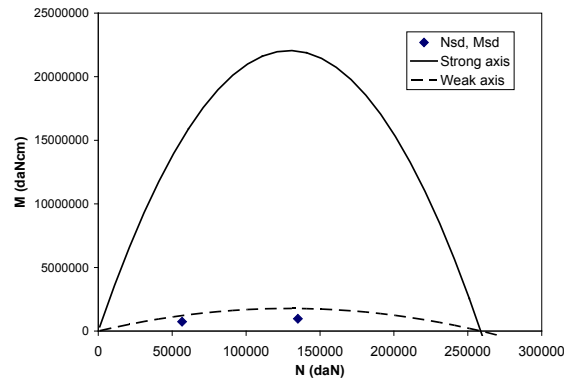


Fig. 26. Flexure-axial force domain of masonry.

The equations that provide the domain are those at translation and rotation in the form

$$a = \frac{N}{t \cdot k \cdot f_u} \quad (7)$$

$$M_{RD} = \frac{N}{2} \cdot (l - a) \quad (8)$$

$$M_{RD} = \frac{N}{2} \cdot l \cdot \left( 1 - \frac{N}{l \cdot t} \cdot \frac{1}{k \cdot f_u} \right) \quad (9)$$

As  $N$  changes, the domain of the side section is calculated. The thickness  $t$  and resistance are obtained using  $k = 0.85-1$ .

Fig. 26 shows the domain of interaction of the cross-section of the stack loaded on and off the plane. In the same graph, the stresses due to calculated values of normal and flexure stresses,  $N_{sd}$  and  $M_{sd}$ , are reported. For calculation of the resistance the values  $L = 2L = 682$  cm,  $t = 55$  cm,  $f_u = 28 / (3 \times 1.35)$  were adopted.

The out-of-plane flexure and normal stresses are only satisfied for the configuration of train load C3, which is the real train that passes in the line. The safety factor for this condition is over 1; consequently, the verification is not satisfied.

## 9. Conclusions

In this paper, the risk of failure of the Salso River railroad steel bridge is analyzed. It has a total length of 112 m and is constituted by three isostatic, reticular beams having a Pratt type composition.

The substructures (two shoulders and two intermediate piles) are in masonry with a reinforced concrete beam in the upper portion.

From visual inspection, it emerges that the structure is affected by corrosion of steel (rivet connections and steel elements). To determine the mechanical characteristics of the materials that constituted the bridge, the piers and the shoulders, non-destructive and partially destructive tests were carried out regarding the following: - a complete geometric survey with careful detection of the crack patterns; - a georadar survey on a flat and inclined surface of the pier; - pacometric tests on upper concrete beams to determine the position and the actual diameter of the reinforcement bars; - endoscopic tests to examine the morphology, typology, and state of conservation of the materials of piers; mechanical characterization of masonry constituting pier through the extraction of long cores specimens to be tested in compression, chemical and mechanical characterization of the mortar; - microhardness tests for estimating the mechanical tensile strength of the steel constituting the bridge deck.

Static verification of steel bridges under permanent and accidental loads utilized actually in the Italian Code was carried out. Verifications of the S.L.U. for steel structures are divided into resistance checks, local stability checks, and global instability checks. For masonry piles, a bending moment-axial force domain was calculated. The results obtained showed that for this bridge, there are no risks of failure and stresses in main members are lower than strength of materials such as for compressed steel strut and masonry piers.

## Declaration of Competing Interest

The authors declared that there is no conflict of interest.

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